

RESEARCH ARTICLE | JUNE 11 2024

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AIP Conf. Proc. 3100, 040013 (2024)

<https://doi.org/10.1063/5.0209472>



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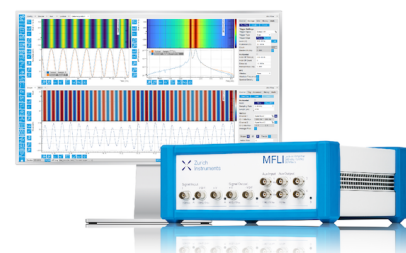
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A Comprehensive Review of Wireless Body Area Network in Medical Applications

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Abstract. In order to improve early diagnosis, prompt intervention, and remote patient care, Wireless Body Area Networks (WBANs) are crucial for the healthcare industry. WBANs provide real-time monitoring of patients' vital signs. Physiological parameters are monitored by small, wearable devices or sensors that make up a wireless body area network (WBAN). WBANs are used in the military, sports, entertainment, healthcare, and entertainment industries. They allow for ongoing observation, remote data access, and better patient care. Interference, power use, data encryption, and user identification are obstacles nonetheless. The development of algorithms, protocols, and hardware solutions to improve WBAN performance and security is the subject of ongoing research. WBANs have a bright future since they could revolutionize environmental interaction and health monitoring.

Keywords. Health Monitoring, patient monitoring, Wireless Body Area Network (WBAN), WBAN security.

I. INTRODUCTION

Wireless Body Area Networks (WBANs) are critical for enabling continuous, real-time monitoring of people's vital signs and health data, as well as providing rapid identification of abnormalities, personalised medical insights, and remote patient care, hence improving healthcare outcomes and resource efficiency. Wireless Body Area Network (WBAN) comprised of tiny, low-power sensors that are implanted or attached to human bodies to monitor physiological parameters and wirelessly transfer the data to a remote device for additional processing or analysis. Medical professionals can now monitor patients' vital signs such as heart rate, blood pressure, temperature, and glucose levels in real-time through WBAN's technology. WBAN technology can be utilised in the sports, fitness, and military industries to track athletes' performance or soldiers' health during drills or battle [2].

WBAN devices deliver data to the distant device using a variety of wireless communication protocols, including Bluetooth, Zigbee, Wi-Fi, and cellular networks. In order to ensure comfortable and inconspicuous wearability, the gadgets are often intended to be lightweight, low power, and have a compact form factor [3]. Despite the potential advantages of WBAN, there are also a number of issues that must be resolved in order to assure its successful adoption in a variety of applications, including energy usage, security, and interoperability. The WBAN can be implanted in different ways. Some common type of WBAN are:

- a. **Implanted WBAN:** This kind of WBAN involves implanting wireless sensors or devices inside the body of a person to track physiological variables including blood pressure, glucose levels, and heart rate. Medical devices like pacemakers, glucose sensors, and brain implants all make use of implanted WBANs.
- b. **On-Body WBAN:** This kind of WBAN uses wireless sensors or devices attached to the surface of the human body to track physiological variables. Applications like fall detection, sports and fitness tracking, and sleep monitoring make use of on-body WBANs.
- c. **In-Body WBAN:** In this kind of WBAN, wireless sensors devices are placed inside body cavities like the gastrointestinal tract or the bladder. Medical uses for in-body WBANs include monitoring gastrointestinal problems and bladder issues.

. Hybrid WBAN: This kind of WBAN combines two or more WBAN types, such as implanted and on-body WBANs, to track a variety of physiological data. Applications like remote patient monitoring and continuous glucose monitoring use hybrid WBANs.

d. Ambient WBAN: In this kind of WBAN, the patient's environment, such as the home or hospital, is equipped with wireless sensors or devices that are used to track physiological indicators. Environmental monitoring and eldercare monitoring are two examples of applications that use ambient WBANs.

WBAN provides a range of features and applications, including:

- 1) Health monitoring: Heart rate, blood pressure, respiration rate, and temperature may all be continuously monitored in real-time using WBAN. In order to detect health issues and provide prompt medical action, WBAN is useful.
- 2) Physical activity monitoring: Individuals' physical activity, including steps taken, miles travelled, and calories burned, can be tracked using WBAN.
- 3) Environmental monitoring: It can be used for environmental factor such as air quality and temperature in real time.
- 4) Emergency response: In order to notify emergency services about medical situations like heart attacks or strokes, WBAN might be employed.
- 5) Chronic disease management: WBAN can be used to track long-term illnesses like diabetes and give patients feedback to assist them control their disease.
- 6) Sports performance monitoring: WBAN can be used to monitor and track an athlete's performance during practice and competition, giving athletes information about their physical state and assisting them in performing better [4].

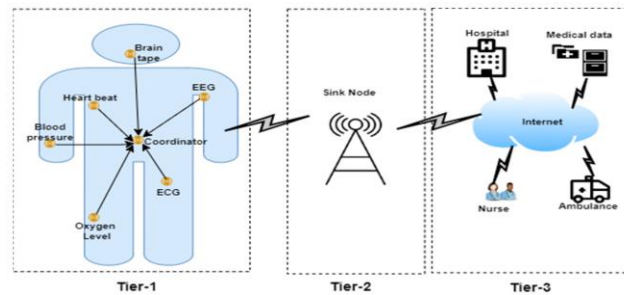


Fig.1. The architecture of wireless body sensor networks

II. DESIGN FLOW / PROCESS

A three tier heterogeneous wireless body sensor network is primarily suggested for indoor patient condition monitoring as shown in Fig 1. Every tier is important for transfer of information.

Tier-1: The sinkhole is placed at this level, and there are medical sensor nodes implanted on or in the human body.

Tier 2: Data are sent from sensor nodes to the sinkhole, where they are processed and aggregated before being sent to the BS.

Tier 3: The data are sent to the medical Centre through the Internet infrastructure for remote monitoring and treatment after being received by the BS. According to the architecture of this kind of network, each sensor generally keeps track

of a patient's health by gathering sensory data from the patient's body and transferring it to a sink node before being sent to the BS to request medical attention [5].

Working Process: The working process of WBANs is shown in Fig 2. The following steps are included in a wireless body area network (WBAN) operational process:

- 1) *Sensor Data Acquisition:* Physiological information, such as heart rate, temperature, or blood pressure, is gathered by the sensors that are affixed to or inserted into the body. Different sensing methods, including ECG, PPG, OR MEMS, may be used by the sensors to collect the data [6].
- 2) *Sensor Data Processing:* The sensors themselves or an MCU that is built within the sensors process the collected data. Filtering, amplifying, or digitizing the data are some examples of processing.
- 3) *Wireless Communication:* Using a wireless communication technology like Bluetooth, ZigBee, or Wi-Fi, the processed data is wirelessly sent to a central device, such as a smartphone or a hub. Real-time gearbox options are available, as well as local storage prior to the subsequent gearbox [7].

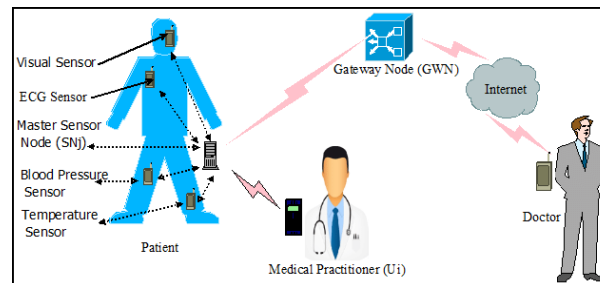


Fig.2. WBAN Working Process

- 4) *Data Storage and Management:* The data is received by the central device, which either keeps it locally or transfers it to a distant server for additional processing and analysis. Either a database or a cloud-based platform could be used to manage the data [7].
- 5) *Data Analysis and Visualization:* To extract useful information, such as trends, patterns, or anomalies, the stored data is analysed. To make the information more comprehensible and usable, it can be visualised using graphs, charts, or other visualisation techniques.
- 6) *Application:* Applications for the analysed data include remote medical diagnosis, tracking athletic performance, and health monitoring. Based on the data, the application can produce warnings, alarms, or recommendations.
- 7) *Security and Privacy:* It is essential to maintain the security and privacy of the sensor-collected data because it is frequently sensitive. The data may be secured using encryption, authentication, or other security measures to prevent unauthorized access or disclosure [8].
- 8) *Power Management:* The sensors typically have a tiny surface area and are mounted on the body or inside it. In order to reduce power consumption and increase battery life, power management techniques like energy harvesting, low-power modes, or duty cycling may be used.

TRENDS IN WBAN RESEARCH

On WBANs, a lot of study is being done. The key concerns include network size, accuracy of results, node density, power supply, mobility, data rate, energy consumption, quality of service, and real-time communication. Due to their small size, WBAN nodes need miniature batteries. In order to maximize the life of power sources, the network must operate and perform in a way that is energy efficient [8].

The majority of the effort in this area has focused on improving MAC protocols for processing that uses less energy [9]. Currently, there are two different methods for creating MAC protocols for sensor networks. The first one is MAC protocol design based on contention. The CSMA/CA (Carrier Sense Multiple Access - Collision Avoidance) MAC protocol is an illustration of this kind of MAC protocol. Priority channel access is given to the nodes in this configuration before data transmission [11]. Almost no time synchronization limitations, simple network variation adaptation, and scalability are advantages of CSMA/CA based protocols. Schedule based MAC protocol is the alternate strategy. A TDMA-based protocol is an illustration of this kind of protocol, which offers time-slotted access to the channel. Thus, various users are given their own time slots for data transfer. These time windows may have set or variable lengths [11]. Time slots are provided by the Time Slot Controller (TSC). According to, IEEE 802.15.4 is unable to offer WBAN applications energy-efficient connectivity [12].

For their usage in the medical field, there are numerous gaps that must be filled. Thus, IEEE 802.15 with task group 6 (BAN) has been developed as a new, better standard [13]. New MAC and physical layers for WBAN are provided by this standard. Narrowband, Ultra-Wide Band (UWB), and Human Body Communication (HBC) are the several frequency bands that make up the physical layer [14].

Another unique MAC approach was presented by Timmons et al. called MedMAC. The protocol makes use of sleep modes, an innovative low-overhead TDMA synchronisation mechanism, energy-efficient time slots, optimal contention periods, and time slots that may change dynamically. It also includes a contention-free TDMA channel access strategy. By employing beacons, the Adaptive Guard Band Algorithm (AGBA) keeps devices in sync when they are sleeping. The protocol only benefits low- and medium-data rate medical applications [14].

For MAC layer processing, an algorithm using cross-layer fuzzy rules was devised by Otal et al. for replacing the traditional first-come, first-served transmission method [15]. In order to meet the requirements for packet transfer reliability and message delay (reliability & message delay) for Distributed Queuing Body Area Networking (DQBAN), a novel DQBAN MAC protocol was developed. Although not much, energy usage was decreased [16].

Cascading Information by Controlling Access with Distributed Slot Assessment (CICADA), a cross-layer protocol based on tree architectures, was also suggested by Latre et al.[18] for use with WBAN. CICADA-S protocol is an expansion of CICADA, a cross-layer protocol that manages data routing in WBANs as well as medium access [19]. The CICADA-S protocol is the first complete method for dealing with dangers in this situation of mobile medical monitoring. Low Health Technology combined a control cycle with a data sub-cycle to reduce delays and increase energy efficiency [20].

IV. ADAPTIVE WBAN APPLICATION USING A CRT

The proposed design combines standard WBAN with cognitive radio technology (CRT), two of the most significant technologies now available, to produce a highly reliable and effective WBAN architecture that will revolutionise the field of wireless body area networks in the future. Spectrum management, networking effectiveness, and energy effectiveness can all be significantly improved with the unique cognitive approach based WBAN [21]. The proposed architecture of WBAN with cognitive capabilities is part of the research design. The three layers of communication in the architecture are as follows:

Level 0: The intra-WBAN communication is part of this level. These elements are included in this:

- *Core and actuator nodes (sensors/actuators)*

Numerous sensor nodes operate on, within, or very near (less than 2 cm) the body to collect data on physical signals (bio-signals) emanating from the human body (detection and acquisition), analyse them (if required), and then communicate them to a centralised body network controller. It is composed of a transceiver, a power unit, and a tiny processor with memory [22].

- *CRT based Network Coordination Unit (NCU)*

The patient's body or the subject of the inquiry itself is where this device is placed. After processing data, the NCU's main job is to transmit it to the Central Controlling Unit (CCU) at Level 1 through corresponding with access and core nodes (sensors/actuators). CRT-based NCUs are capable of a variety of cognitive tasks, including data fusion, spectrum sensing (between nodes and NCU), configurable networking, node parameter sensing (mobility, battery life, working, connectivity), energy-efficient and reliable data routing, and wireless cognitive connectivity with CCU [23]. A CRT-based NCU would consist of a transceiver, a small-size processor unit, and a memory unit as its three main components.

Level 1:

This level of communication is inter-WBAN networking. Here, a number of NCUs are connected wirelessly to the Central Controlling Unit (CCU), which is equipped with CRT technology. The main responsibility of the CCU is to use CRT technology to connect with a number of NCUs located inside a certain space (such as a hospital, building, or institution) [20]. Power supply and connectivity difficulties are less crucial because this machine will be static, but the other issues are quite significant. Cognitive functions that can be incorporated in CRT enabled CCU include spectrum sensing (between NCUs and CCU), energy-efficient and highly reliable routing protocols, efficient MAC protocol, NCU parameter maintenance (mobility & location, battery life, working, calibration and connectivity), QoS awareness, and end-to-end goal maintenance [24].

Level 2

This level of communication goes beyond WBAN networking. Any of the following such as a doctor, medical assistant, or member of the support staff, Emergency services, such as an ambulance or risk management personnel and Medical Facility or Care Unit may be wirelessly connected to a CRT-based CCU.

CONCLUSION

Wireless body area networks (WBANs), a promising medical technology for remote monitoring and personalised care, have grown in popularity. WBANs are tiny, low-power sensors that are implanted, attached to things, or put around patients to gather physiological data and wirelessly transfer it for analysis. The fundamental objective of WBANs is to improve the caliber of healthcare services by enabling early diagnosis and intervention as well as real-time monitoring of vital signs.

WBANs are used in the healthcare industry to continuously monitor patients' vital signs, allowing medical professionals to monitor patients' situations remotely and provide quick medical assistance. WBANs help track athletes' athletic and fitness progress and provide recommendations for enhancement. In order to protect the safety of soldiers, WBANs are used in military applications to monitor soldiers' health during combat or training.

The creation of wireless body area networks has the power to revolutionise healthcare and improve people's quality of life. By overcoming obstacles and exploiting developments in sensor technology, communication protocols, and data processing techniques, WBANs can contribute to the provision of more personalised and proactive healthcare. This will make it possible to detect diseases early, monitor patients from a distance, and improve overall health.

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